

Markscheme

Mathematics: analysis and approaches

Practice question bank

82. (a)

A car initially worth \$28 000 depreciates at 15% per year.

Find the number of complete years after which the value of the car first falls below \$10 000.

$$\text{solve } 28\,000(0.85)^n < 10\,000: (0.85)^n < 0.357\dots \quad (M1)$$

$$n > \frac{\ln(0.357\dots)}{\ln(0.85)} = 6.34\dots \quad (M1) \text{ A1}$$

check $n = 6$: value $\approx$ \$10 560 (still above); check $n = 7$: value $\approx$ \$8976 (M1)

first complete year below \$10 000 is $n = 7$ A1

[5 marks]